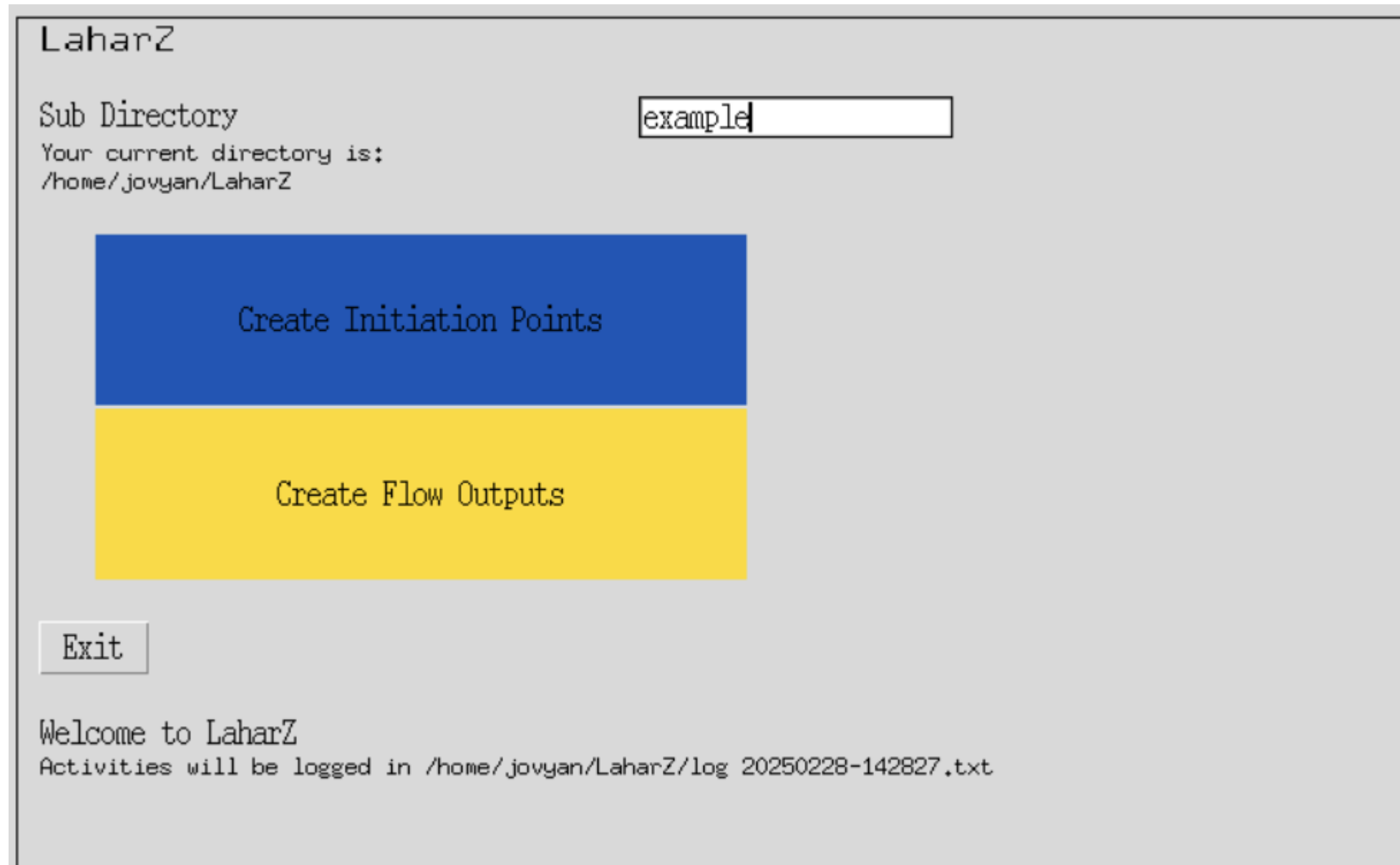


LaharZ Example

A) Model setup

1. Login to VICTOR through `hub.victorproject.org` in your browser
2. Choose a small machine
3. Once logged in, open a Terminal window, type `victor setup` and choose LaharZ
4. Copy the DEM of Mt Rainier named `Rainier.tif` from `~/shared/DEMs/NorthAmerica/` into the new folder `LaharZ/example`
5. Launch the Desktop using the  icon
6. Open a Terminal window on the desktop by clicking the  icon
7. Change directory into the folder `LaharZ` by typing `cd ~/LaharZ` in the terminal window and pressing enter.
8. Run the program by typing `python laharz.py` and pressing enter. A new window will appear.

In the GUI, type “example” for the folder name, and click **Create Initiation Points**



The screenshot shows a terminal window titled "LaharZ". Inside, there is a "Sub Directory" label followed by a text input field containing the word "example". Below this, it says "Your current directory is: /home/jovyan/LaharZ". In the center, there are two large rectangular buttons: a blue one labeled "Create Initiation Points" and a yellow one labeled "Create Flow Outputs". At the bottom left, there is a small "Exit" button. At the bottom of the window, a welcome message reads "Welcome to LaharZ" and a note states "Activities will be logged in /home/jovyan/LaharZ/log 20250228-142827.txt".

LaharZ

Sub Directory

Your current directory is:
/home/jovyan/LaharZ

Create Initiation Points

Create Flow Outputs

Exit

Welcome to LaharZ
Activities will be logged in /home/jovyan/LaharZ/log 20250228-142827.txt

A) Creating initiation points:

1. Enter the input parameters for the initiation points

1. Click the “Initiation points” button

1. If asked, click “overwrite” button

Generate Initiation Points

DEM File

Rainier.tif

Name of your DEM file in example

Stream and Flow Direction Files

• Generate files

• Use your own files

☒ Fill DEM file

Streams file

☒

MR-streams.tif

Name of your streams file to be created in example

Stream Threshold

1000.0

Threshold value for accumulation to form stream

Flow direction file

☒

MR-flowdir.tif

Name of your flow direction file to be created in example

Determine Energy Cone Apex

• Use Search File

• Use Longitude/Latitude

Apex

-121.75, 46.85

Longitude, Latitude

Incremental Height

50.0

Incremental height in metres

H/L Ratio

0.4

H/L Ratio

Sea Level

0.0

No initiation points will be created at sea level or below

☒ Plot Energy Cone Graphics

Energy Cone Graphics File

☐

MR_ec_cone.tif

Name of the file for the energy cone graphics in example

Extent

1.5

Extent to plot the energy cone/surface (1.3 = 130% of L)

☐ Energy Cone Line File

☐ Proximal Hazard Zone File

☐ Initiation Points File

MR_ecline.tif

prox_hz_zn.gpkg

MR_ip1.gpkg

Name of the file for the energy cone line in example

Name of the file for the proximal hazard zone in example

Name of the file for the initiations points in example

Back

Initiation points

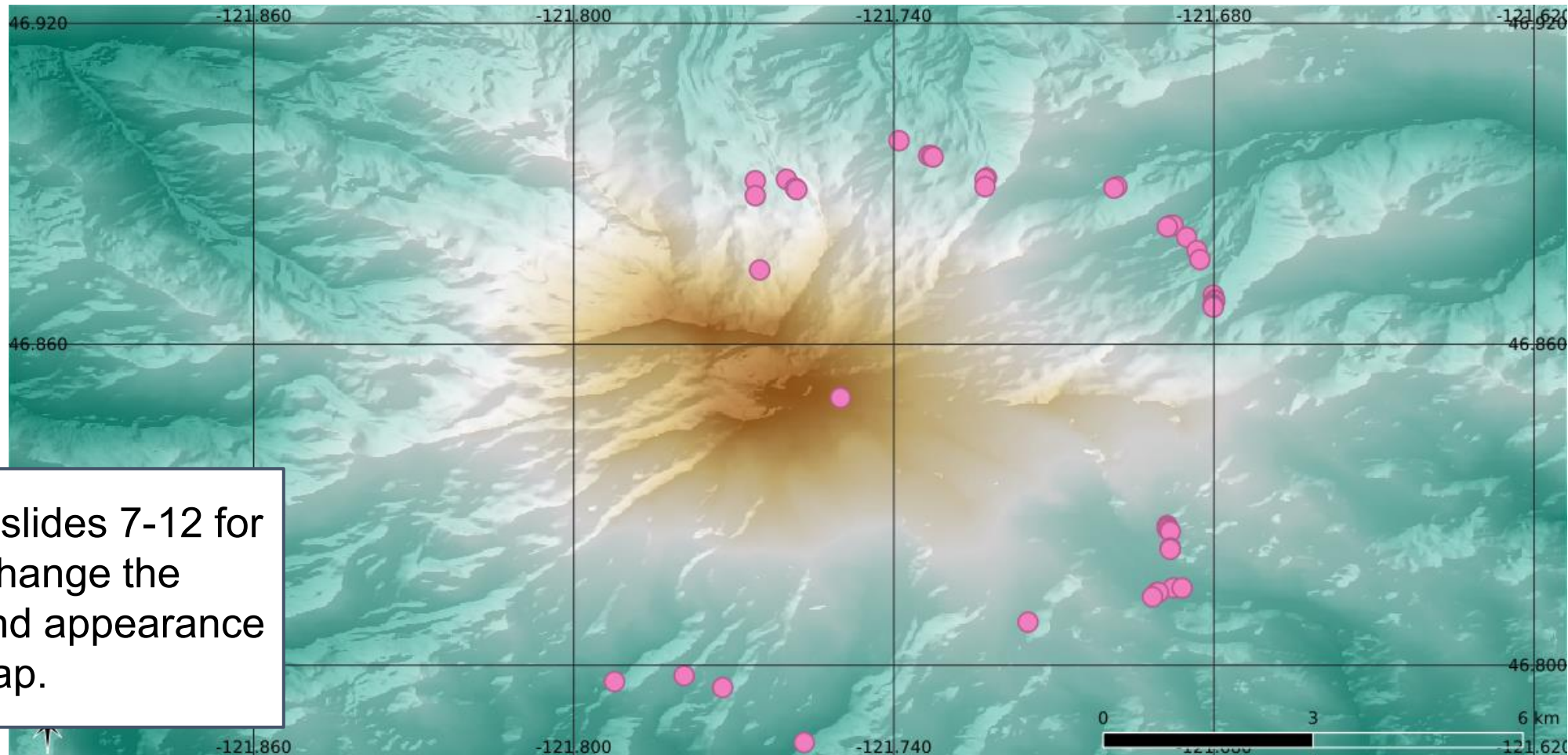
Visualization of Initiation Points

1. On the Desktop, open QGIS



2. Load the Mt Rainier DEM using “Layer->Add Layer->Add Raster Layer”

3. Add the initiation points just created using “Layer->Add Layer->Add Vector Layer” and loading the file MR_ip1.gpkg



! See slides 7-12 for how to change the colors and appearance of the map.

B) Run the flow model

1. In the GUI, click and run **Create flow outputs**
2. Enter the file names to use the results of the first step
3. Update flow parameters:
 - o Change the Scenario to “Custom” and the Planar Area to 200.0 (note the .0)

Generate Flows

DEM File

Rainier.tif

Name of your DEM file in example

Streams File

MR-streams.tif

Name of your streams file in example (not used to generate flows)

Flow Direction File

MR-flowdir.tif

Name of your flow direction file in example

Initiation Points File

MR_ip1.gpkg

Name of the file with the initiations points in example

Volume(s)

1e4, 1e5, 5e5

Volumes (m³) in a list separated by commas

Scenario

Custom

Name of the scenario to use

Cross sectional area:

c 1

0.05

Value of the c1 parameter

Planar Area

c 2

200.0

Value of the c2 parameter

Cross Sectional Area =

c 1 V

$\frac{c_1 V}{c_2}$

Planar Area =

c 2 V

$\frac{c_2 V}{c_1}$

Sea Level

0.0

Flow will stop at sea level

Flow Output Directory

☐

MR_Lahars

Directory for flow output raster files

Flow output vector file

☒

File name for flow vector file

Back

Create Flows

Volume(s)

1e4, 1e5, 5e5

Scenario

Lahar

Lahar
Debris
Pyroclastic
Custom

Cross sectional area:

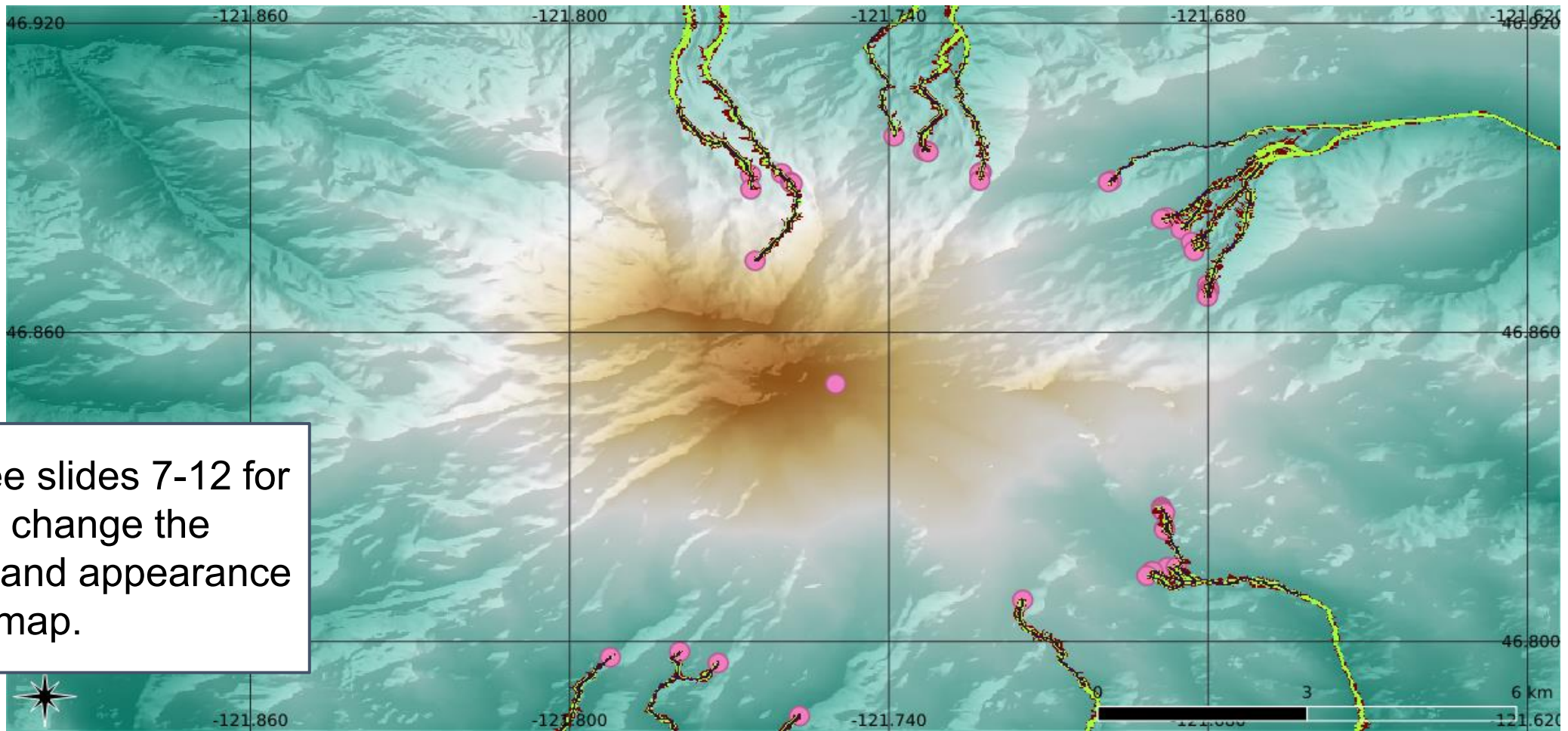
c 1

Planar Area

c 2

Visualization of flow outputs

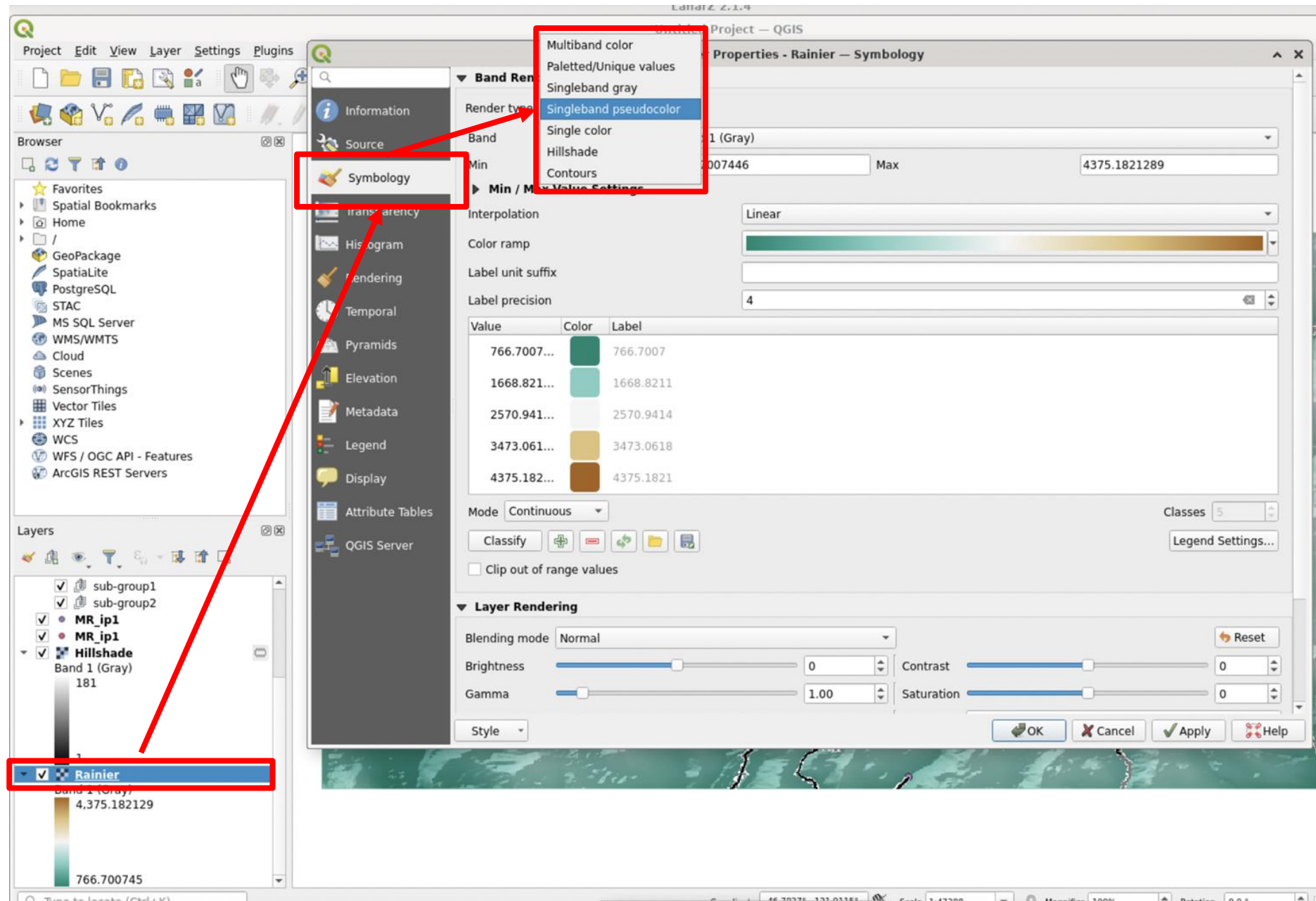
1. Return to your QGIS project
2. Add flow output tiff files from the results folder using Layer→Add Layer →Add Raster Layer and choosing the file `total.tif` from the MR_Lahars folder



! See slides 7-12 for how to change the colors and appearance of the map.

C) Adjust the visualization of the flow outputs: color

1. In the 'layers' panel at the bottom left of your QGIS screen, double click the Rainier layer.
2. A window will appear. From here, ensure you are in the 'symbology' tab from the list on the left of the window.
3. Change the drop down menu labeled "Render Type" at the top of the window to "Singleband Pseudocolor"



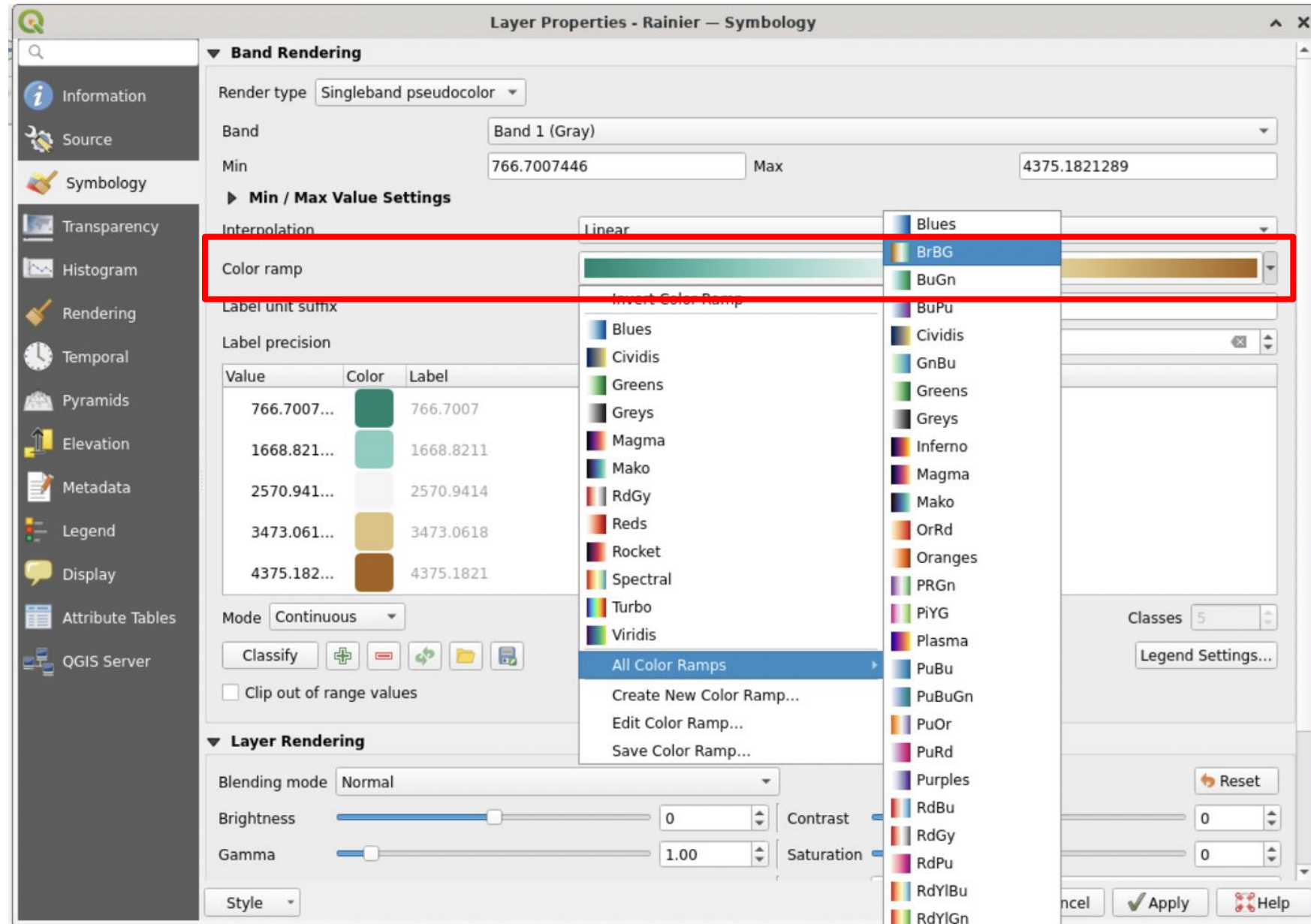
c) Adjust the visualization of the flow outputs: color

4. In order to match the example, click the arrow to the right of the 'color ramp' option.

5. Hover over 'All color ramps' from the list that appears and choose "**BrBG**."

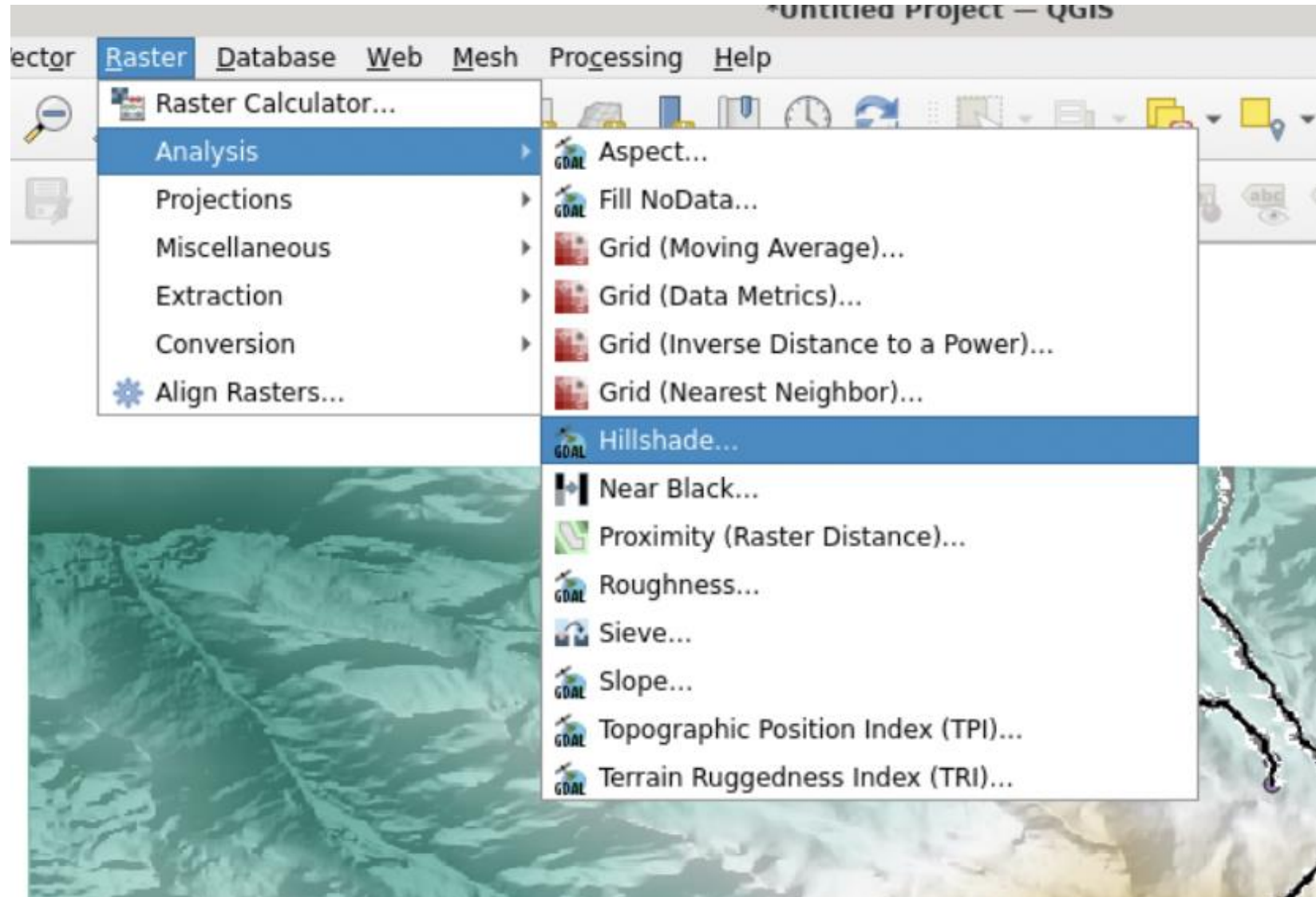
6. Click the arrow to the right of color ramp again and click 'Invert Color Ramp' at the top of the list that appears.

7. Click apply before closing the window.



d) Adjust the visualization of the flow outputs: topography

1. Navigate to the Raster menu item at the top of the window.
2. Hover over 'Analysis' and select 'Hillshade' from the list that appears.
3. After ensuring the DEM being used is `Rainier.tif`, select 'run' at the bottom of the window that appears.

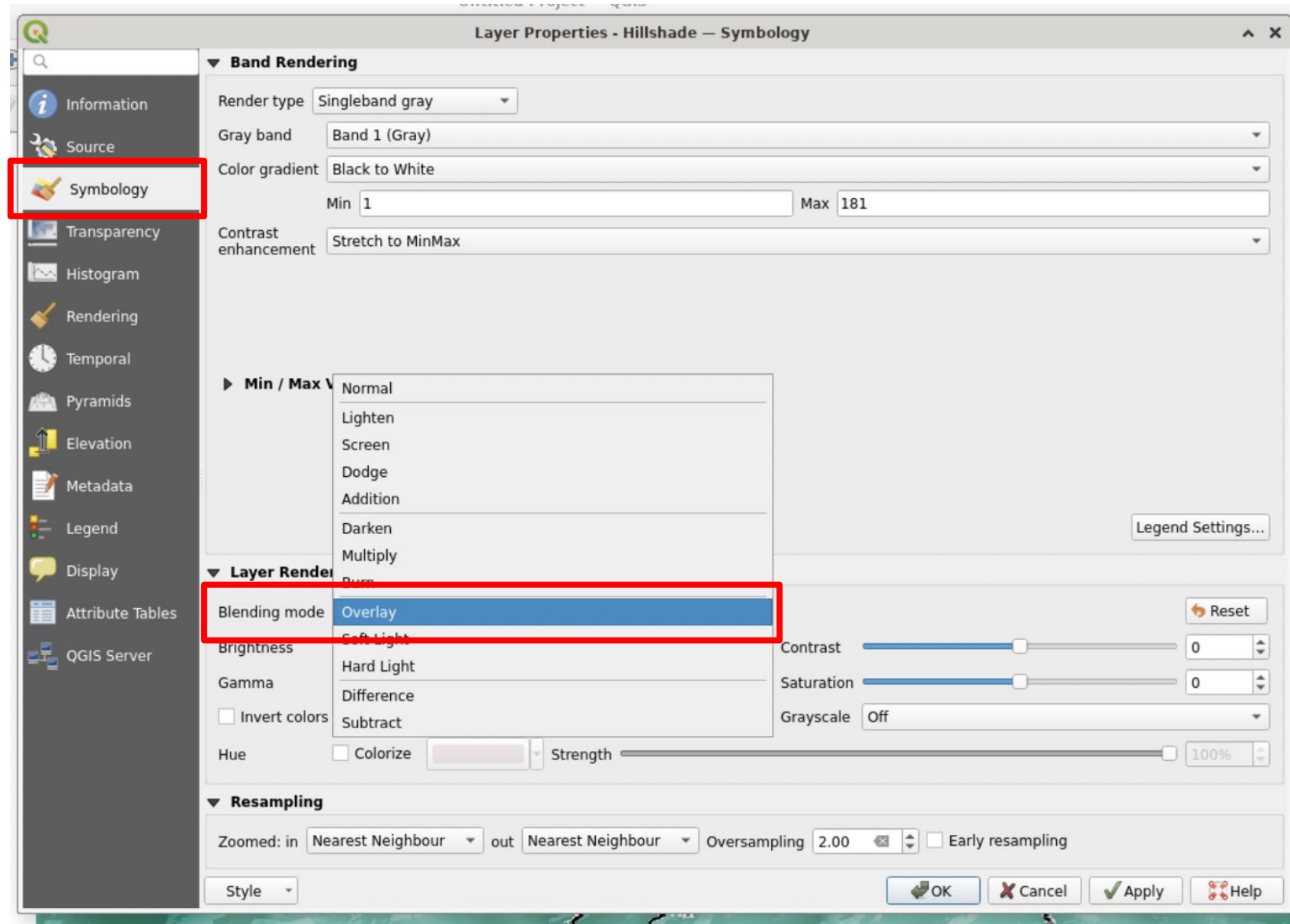


d) Adjust the visualization of the flow outputs: topography

4. Close the window and double click the new 'Hillshade' layer in the layers panel.

5. Ensure you are in the symbology tab. Open the dropdown menu under the "Blending Mode" option and select overlay.

6. Click apply at the bottom of the window



d) Adjust the visualization of the flow outputs: topography

7. Open the 'Transparency' tab in the same window.

8. Adjust the 'global opacity' slider to around 25%.

9. Click apply at the bottom of the window.

